

Git & GitHub Setup Guide

Software Engineering Course: Quick-Start Guide for Students

1 Installing Git

Git is the tool that tracks changes in your code. Install it based on your operating system:

Windows

- Go to git-scm.com/download/win, the download starts automatically.
- Open the downloaded **.exe** file and click **Next** through the setup (default options are fine for beginners).
- Once installed, open **Git Bash** from the Start Menu.

Mac

- Open **Terminal** and type the command below, then press Enter:

```
xcode-select --install
```

- This installs Git along with Apple's Command Line Tools. If Git is already installed, you'll see a message saying so.

Note: Alternatively, Mac users with Homebrew can run: `brew install git`

Linux

- Open your Terminal and run the command matching your distribution:

```
sudo apt update && sudo apt install git (Ubuntu/Debian)
```

```
sudo dnf install git (Fedora)
```

Verify the Installation (all OS)

- Open Terminal / Git Bash and run:

```
git --version
```

- If you see a version number (e.g. `git version 2.44.0`), Git is installed correctly.

2 Creating a GitHub Account

Use your **college email ID** (the one ending in your institute's domain) while signing up. This is required later to verify your student status for GitHub's free Pro benefits.

- Go to github.com/signup.
- Enter your **college email address**, create a password, and choose a username.
- Verify your email using the code GitHub sends you.
- Complete the short setup questions (these just personalize your experience, any answer is fine).

Note: Already have a personal GitHub account? You can still create a separate one with your college email, or simply add the college email to your existing account under Settings > Emails. Either approach works for Section 4.

3 Transferring a Repo to Your College Account

If you already have a project on your personal GitHub account and want to move it under your college account, transfer it instead of re-uploading it. This keeps all your commit history intact.

- Open the repository on your **personal** account and go to **Settings**.
- Scroll to the bottom of the page to the **Danger Zone**.

- Click **Transfer ownership**.
- Type the repository name to confirm, and enter your **college account's username** as the new owner.
- Click **I understand, transfer this repository**.
- Accept the transfer from your college account. You'll get a notification/email once it's ready.

Note: This is optional, only needed if you want an existing project to live under your college account.

4 GitHub Pro + Copilot (Student Developer Pack)

Make sure you are logged into your **college-email account** before starting. Complete these four steps in order:

Step 1: Turn on Two-Factor Authentication (2FA)

Settings > Password and authentication > Two-factor authentication

- Under **Preferred 2FA method**, choose **Authentication app**.
- Follow the on-screen steps to scan the QR code with an authenticator app (e.g. Google Authenticator) and enable it.

Step 2: Add Billing Information

Settings > Billing and licensing > Payment information

- Add a valid card/billing address in this section.

*Note: This is only for verification. You will **not** be charged anything. Your application won't be approved without this step.*

Step 3: Apply for Educational Benefits

Settings > Billing and licensing > Educational benefits

- Click **Apply** and fill in your school details.
- Upload your proof of enrollment (college ID card or enrollment letter) when asked.
- Submit and wait. GitHub will review and send a confirmation email once approved (usually within a few days).

Step 4: Set Up Copilot

- Once you receive the approval email, go back to **Settings > Billing and licensing**. Your account will now show **GitHub Pro** and **Copilot** as available.
- Enable Copilot from there, then install the **GitHub Copilot** extension in VS Code (or your editor) and sign in with the same account to start using it.